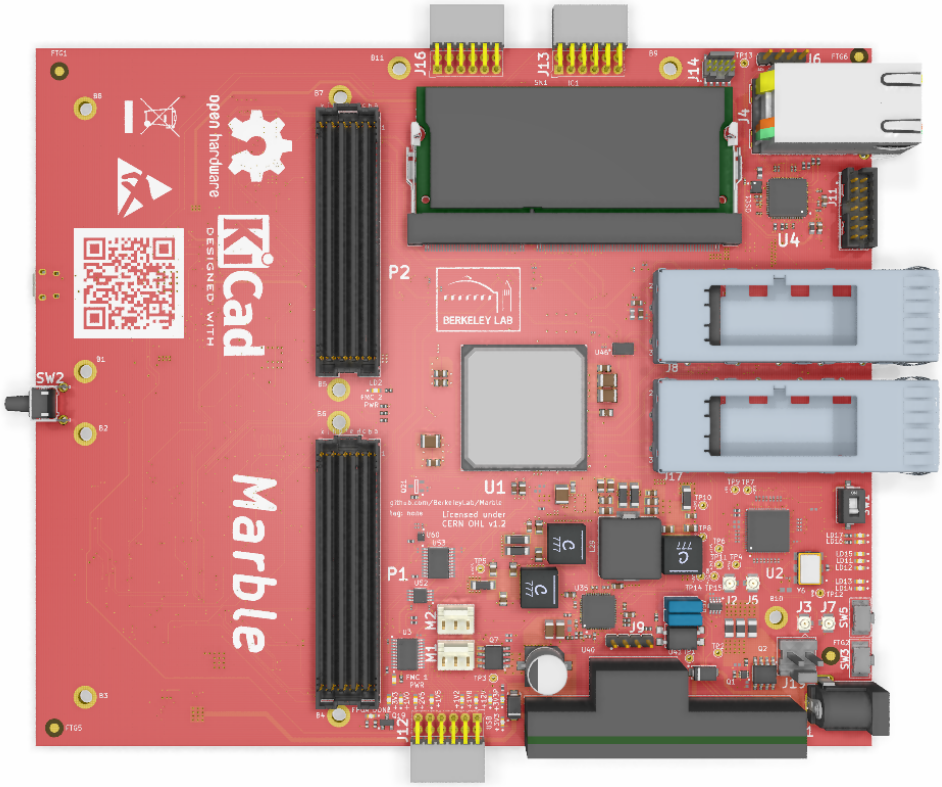
Marble – User Guide

v1.0 2021

February 3, 2022



Introduction

Marble is a fully open source dual FMC carrier board designed for the Accelerator Technologies Group of the Engineering Division of Lawrence Berkeley National Laboratory. This document presents the technical documentation of the Marble module divided into individual functional sections. Design files are made in KiCad and are licensed under the CERN OHL v. 1.2.

Contents

1 Overview **3**
- 1.1 FPGA reset . . . 7
- 1.2 JTAG . . . 7
  - 1.2.1 External JTAG . . . 7
  - 1.2.2 Internal USB-JTAG . . . 7
  - 1.2.3 Self-JTAG . . . 8
- 1.3 LEDs . . . 8
- 1.4 FPGA Programming . . . 8
  - 1.4.1 Internal JTAG . . . 8
  - 1.4.2 External JTAG . . . 9

2 SO-DIMM **9**

3 GTH Routing **9**

4 Clocking **10**
- 4.1 White Rabbit clock generator . . . 12

5 Pmod connectors **12**

6 QSFP **12**

7 Ethernet **15**

8 FMC **15**

9 USB-UART **16**
- 9.1 FPGA UART . . . 16
- 9.2 MMC UART . . . 16

10 Power **16**
- 10.1 Fan controller . . . 18

11 MMC **19**
- 11.1 Programing . . . 20
- 11.2 LEDs . . . 21
- 11.3 I2C Tree . . . 21

12 Mechanical dimensions **23**

13 Appendix **26**
- 13.1 Power supply test points . . . 26

1 Overview

> Design files are open source and can be downloaded from GitHub:
> https://github.com/BerkeleyLab/Marble

Marble is a dual FMC carrier module based on an Kintex-7 FPGA. The block diagram of the module is shown in figure 2.

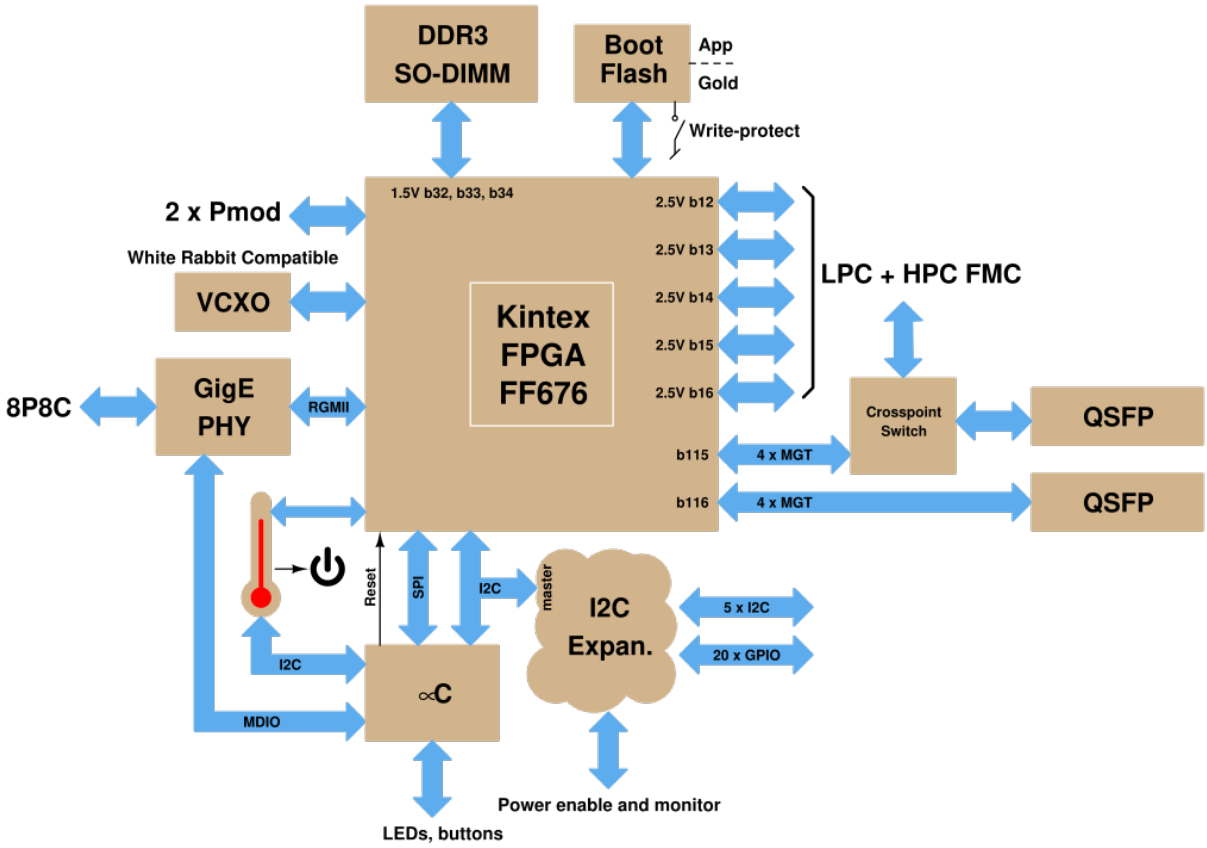


Figure 1: Marble block diagram.

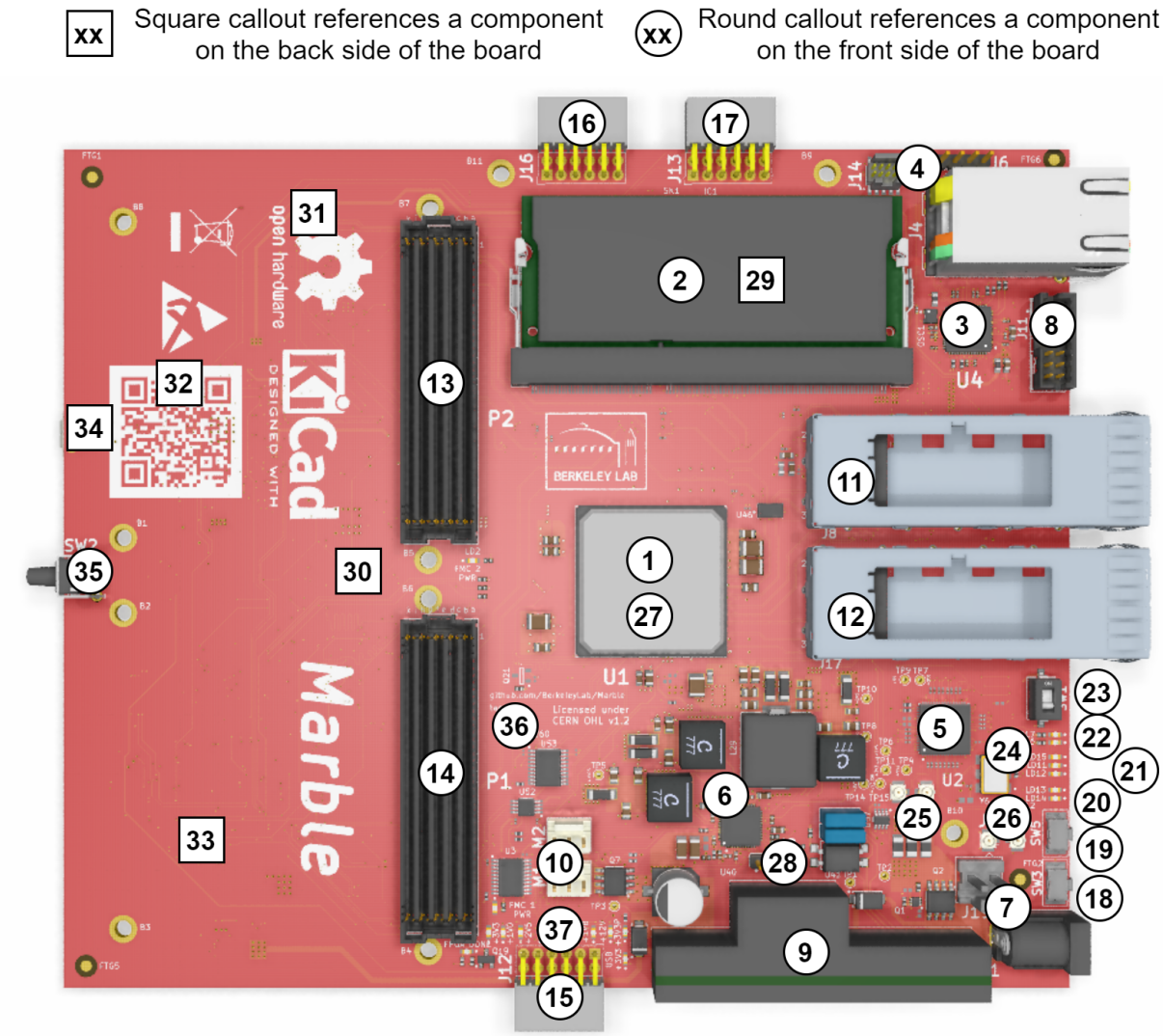


Figure 2: Marble board components.

Callout	Reference Designator	Component Description	Notes	Schematic Page
1	U1	FPGA Xilinx Kintex-7	XC7K160T-2FFG676C	
2	SK1	DD3 SO-DIMM memory module	VR7PU286458FBAMJT	
3	U4	10/100/1000 Ethernet PHY	88E1512-A0-NNP2I000	
4	J14, J6	Microcontroller programing connectors		
5	U2	8x8 Clock Crosspoint Switch	ADN4600ACPZ	
6	U35	Power Management	XRP7724ILBTR-F	
7	J1, J19	12V power input	FC68148(DC-10A), 641119-2	
8	J11	FPGA JTAG connector	87831-1420	
9	U40	PoE module	AG5300	

10	M1, M2	3-pin PC fan connectors	SWR25X-NRTC-S03-ST-BA
11	J8	QSFP connector	QSFP8-038-01-L-D-RA1
12	J17	QSFP connector	QSFP8-038-01-L-D-RA1
13	P2	FMC HPC connector	ASP-134486-01
14	P1	FMC HPC connector	ASP-134486-01
15	J12	PMOD connected to FPGA	PPTC062LJBN-RC
16	J16	PMOD connected to microcontroller	PPTC062LJBN-RC
17	J13	PMOD connected to FPGA	PPTC062LJBN-RC
18	SW3	User button connected to microcontroller	KSS241GLFS
19	SW5	FPGA reset button	KSS241GLFS
20	LD13, LD14	User LEDs connected to shared I2C bus	KPH-1608CGCK
21	LD11-12, LD15	User LEDs connected to microcontroller	KPH-1608CGCK
22	LD16, LD17	User LEDs connected to FPGA	KPH-1608CGCK
23	SW1	Memory write protection switch	A6SN-1101
24	Y6	10-280MHz Clock generator	SI570
25	J2, J5	External clock source input	U.FL
26	J3, J7	External clock source input	U.FL
27	U44-U50	GTX Transceivers Mux	PI3DBS12212AZBSEX
28	J9	Power Management programming header	0.1 inch male 4-pin header
29	Y1-Y3, U20	Internal 125 MHz & 20MHz clock sources	CDCM61004RHBT
30	U30	FPGA SPI flash memory	S25FL128SAGMFIR01
31	U5	I2C multiplexer	TCA9548ARGER
32	U23	Double USB - UART bridge, USB-JTAG bridge for FPGA	FT4232H-56Q
33	U54	Housekeeping Microcontroller	STM32F207VCTx
34	J10	Micro USB connector for U23	10103594-0001LF
35	SW2	Not populated user button (physically connected to SW3)	SKHHLQA010
36	U60	Over-voltage and Under-voltage Reset IC	TPS3703A7330DSERQ1
37	LD4-10, LD18-19	Power rails indicator LEDs	KPH-1608CGCK

Table 1

The board has the following functionalities and features:

1. Xilinx Kintex-7 FPGA XC7K160T-2FFG676C
2. Supports FPGA golden image
3. Housekeeping microcontroller (Module Management Controller) with UART console
4. DDR3 204-SODIMM memory module connector. The board supports up to 4 GB memory.
5. Two FMC HPC connectors, but not all signals are connected to the FPGA.
6. 1Gb Ethernet with PoE
7. Built-in clock generator that supports White Rabbit synchronization
8. Various input clock configurations
9. Two QSFP cages that support data transfer up to 40 Gb/s each
10. Built-in USB JTAG which works with OpenOCD

FPGA Bank	Bank Power Supply	Description
Bank 12 HR	+2.5V	FMC2 LA 00-16 (plus SPI, Self JTAG)
Bank 13 HR	+2.5V	FMC2 HA
Bank 14 HR	+2.5V	FMC2 LA 17-33 (plus config, Pmod)
Bank 15 HR	+2.5V	FMC1 LA 00-16 (plus I2C, UART, Pmod)
Bank 16 HR	+2.5V	FMC1 LA 17-33 (plus RGMII)
Bank 32 HP	+1.5V	DDR3
Bank 33 HP	+1.5V	DDR3 (plus Pmod, White Rabbit)
Bank 34 HP	+1.5V	DDR3

Table 2

The S25FL128SAGMFIR01 configuration memory is connected to the FPGA chip. By default, the FPGA chip loads the configuration from the flash memory after correct power cycle. The module is equipped with a switch (fig 3) that blocks the programming of the configuration memory. When the switch is in the ON position, Write Protection is enabled. WP signal status can be read via I2C IO expander.

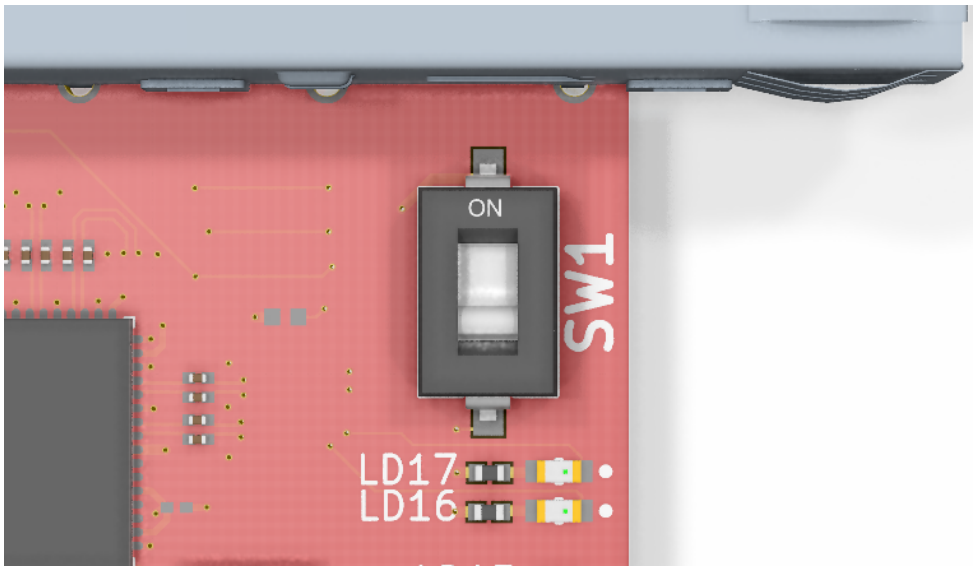


Figure 3: Memory write protection switch.

1.1 FPGA reset

During power system startup, the U60 chip keeps the PROGRAM_B signal low. When a valid supply voltage is detected on the last power-up sequence, U60 changes state on the PROGRAM_B signal which causes the configuration file to be loaded from flash memory.

A local operator can reset the FPGA manually using the SW5 button. Additionally, the MMC can reset the FPGA.

1.2 JTAG

There are 3 sources of JTAG for FPGA:

1. external JTAG (highest priority)
2. internal USB-JTAG (middle priority)
3. self JTAG (lowest priority)

1.2.1 External JTAG

When the external JTAG is connected to J11, GNDDetect signal from the connector switches the multiplexer to pass JTAG signals from the connector to the FPGA. After unplugging cable, the GNDDetect signal is not present and the multiplexer connects internal USB-JTAG to FPGA.

When external JTAG is connected, any other JTAG sources are not available.

1.2.2 Internal USB-JTAG

Internal USB-JTAG is done by using the first data channel of FT4232, which can work as a JTAG. When the micro USB cable is connected, +5V from USB bus switches the multiplexer to pass data from FT4232 to FPGA.

1.2.3 Self-JTAG

Internal self-JTAG can be used only when USB cable and external JTAG are not connected. In this configuration FPGA JTAG signals are connected to FPGA Bank 12:

Signal name	Self JTAG signal	FPGA pin
JTAG TDI	Self_FPGA_TDI	IO_L10P
JTAG TCK	Self_FPGA_TCK	IO_L10N
JTAG TMS	Self_FPGA_TMS	IO_L20P
JTAG TDO	Self_FPGA_TDO	IO_L20N

Table 3

1.3 LEDs

Two general purpose LEDs are connected to the FPGA chip:

1. LD16 - connected to pin IO_L18P_33
2. LD17 - connected to pin IO_25_33

1.4 FPGA Programming

Vivado reference design with a constraint file can be found here:

> https://github.com/BerkeleyLab/bedrock

See its projects/test_marble_family directory.

1.4.1 Internal JTAG

Download the latest version of the FPGA testing code from GitHub:

> https://github.com/BerkeleyLab/Bedrock

> Before testing the FPGA, it is recommended to set up the current limit to 2A on the lab power supply.

Program the FPGA using the following steps:

1. Plug micro USB cable
2. Go to the folder **Bedrock/projects/test_marble_family/**
3. Open command terminal and run command:

```
$ mutil usb
```

4. After the successful programming, LEDs LD16 and LD17 should blink alternately.

1.4.2 External JTAG

Programming FPGA using Vivado and Digilent JTAG HS3 connected to J11:

1. Run Vivado
2. Go to Flow » Open Hardware Manager and then Tools » Auto Connect
3. Click Tools » Program Device » xc7k160t_0 to open the programming window.
4. Choose the *bitstream file* and click Program
5. After the successful programming, LEDs LD16 and LD17 should blink alternately.

2 SO-DIMM

The size of the DDR3 memory can be determined by the user by assembling the appropriate SO-DIMM module to the board. The use of a 204 pin SO-DIMM connector allows up to 4 GB of RAM to be connected to the FPGA. An I2C interface is provided to the memory module so that additional information about the module can be read. The default power supply for memory and HP banks is set to 1.5V and can only be changed by changing resistors. ECC is not supported by the module.

A reference design with a memory controller and a constraint file can be found here:

> https://github.com/BerkeleyLab/bedrock

See its projects/test_marble_family directory.

3 GTH Routing

Gigabit transceivers routing can be configured by the microcontroller. Transceivers from Bank 116 are permanently connected to module QSFP1. Transceivers from Bank 115 can be routed in 3 ways:

1. 4 transceivers connected to QSFP2
2. 4 transceivers connected to FMC P2
3. 2 transceivers connected to FMC P2 and 2 transceivers connected to FMC P1

Transceiver routing configuration can be set by the microcontroller. Three signals control the multiplexers which provide high quality signal switching. Gigabit multiplexer switching can be done from the UART console. By default, the multiplexers are set to route signals to the second QSFP connector. In the table 4, the MUXx columns correspond to the controlling logical state.

MUX3	MUX2	MUX1	MGT4	MGT5	MGT6	MGT7
0	0	0	FCM2-DP0	FMC2-DP1	FMC2-DP2	FMC1-DP1
0	0	1	FCM2-DP0	FMC2-DP1	FMC1-DP0	FMC1-DP1
0	1	0	FCM2-DP0	FMC2-DP1	FMC2-DP2	FMC2-DP3
0	1	1	FCM2-DP0	FMC2-DP1	FMC1-DP0	FMC2-DP3
1	X	X	QSFP2:3/10	QSFP2:1/12	QSFP2:2/11	QSFP2:4/9

Table 4: GTH transceivers routing table

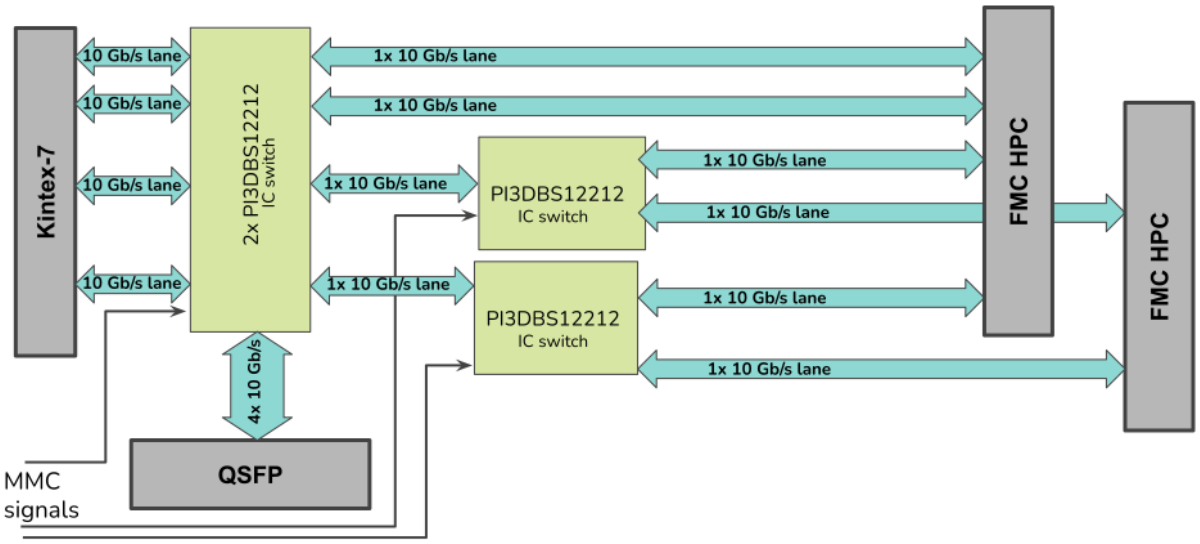


Figure 4: Transceivers routing block diagram.

Use UART console to change the transceiver routing style.

4 Clocking

This section describes how and where clock signals are routed. There are 7 on-board clock sources:

Marble supports external clock sources from FMC and from U.FL connectors. Thanks to the clock multiplexer, any clock input can be connected to any FPGA MGT clock input. Clock routing and Si570 frequency can be changed over I2C from the house-keeping microcontroller or the FPGA.

Clock Name	Reference	Description
System Clock	Y1, Y2	Y1 and Y2 can be soldered interchangeably and provide a reference clock source for the U20 chip
	U20	U20 is an ultra-low jitter clock generator that provides a 125 MHz clock for the FPGA bank 33 that can operate as a DDR3 reference clock. Additionally, U20 generates 125 MHz clock which is connected to the clock multiplexer.
Additional Clock	Y3	VCXO - provides 20 MHz clock for FPGA bank 33
User Clock	Y6	Si570 - provides variable clock frequencies for 10 MHz to 280 MHz; connected to the clock multiplexer
User U.FL Clock 1 (differential pair)	J2, J5	External clock source input
User U.FLClock 2 (differential pair)	J3, J7	External clock source input
FMC1 M2C	P1	GBTCLK 0 & 1
FMC2 M2C	P2	GBTCLK 0 & 1

Table 5

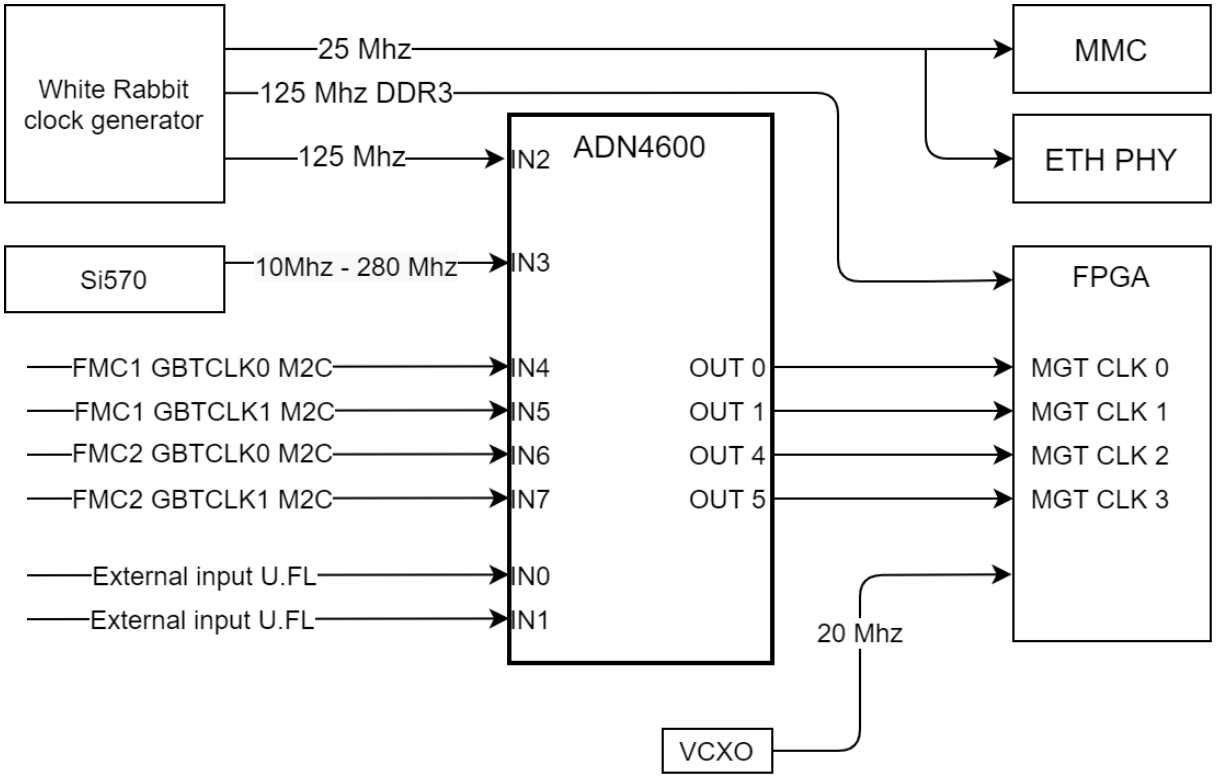


Figure 5: Marble clocking scheme

4.1 White Rabbit clock generator

The clock chips available on the board allow implementation of the White Rabbit protocol. White Rabbit provides sub-nanosecond accuracy and picoseconds precision of synchronization for large distributed systems.

5 Pmod connectors

Marble has 3 Pmod connectors that support 3.3V logic level:

1. J12 and J13 are connected to FPGA
2. J6 is connected the MMC

	Pmod 1 (J12) FPGA	Pmod 2 (J13) FPGA	Pmod (J16) MMC
Pmodx_C_0	IO_L6N_14	IO_L7P_33	PB9 (SEL)
Pmodx_C_1	IO_L7N_14	IO_L2N_33	PC3 (MOSI)
Pmodx_C_2	IO_25_14	IO_L4N_33	PC2 (MISO)
Pmodx_C_3	IO_L7P_14	IO_L7N_33	PB10 (SCK)
Pmodx_C_4	IO_0_14	IO_L2P_33	PB14 (EINT1)
Pmodx_C_5	IO_L5N_15	IO_L8P_33	PB15
Pmodx_C_6	IO_L4P_15	IO_L4P_33	PD6 (UART4 RX)
Pmodx_C_7	IO_L5P_15	IO_L3N_33	PD5 (UART4 TX)

Table 6: Pmod connectors pins assignment

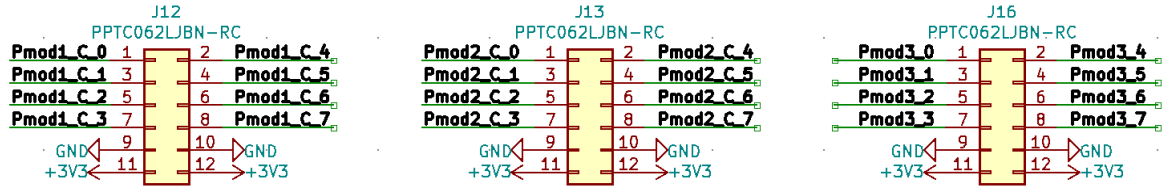


Figure 6: Pmod pinout

6 QSFP

Marble is equipped with 2 QSFP cages (7). Each can support 40 Gb/s of data transfer:

1. QSFP number 1 - high-speed lines are directly connected to the FPGA bank 116
2. QSFP number 2 - high-speed lines are connected to the FPGA bank 115 through the gigabit multiplexer.

QSFPs control signals are connected to the I2C IO expander (U34)(8) which is accessible from FPGA or MMC. The connection of the differential signals to the FPGA is shown in the tables (7)(8). QSFP2 is not directly connected to the FPGA chip. It is connected via a gigabit multiplexer which is controlled from the MMC. By default, the multiplexer is configured to connect the QSFP2 to the FPGA. In order to choose a different configuration or restore the default, select the appropriate option from the MMC console.

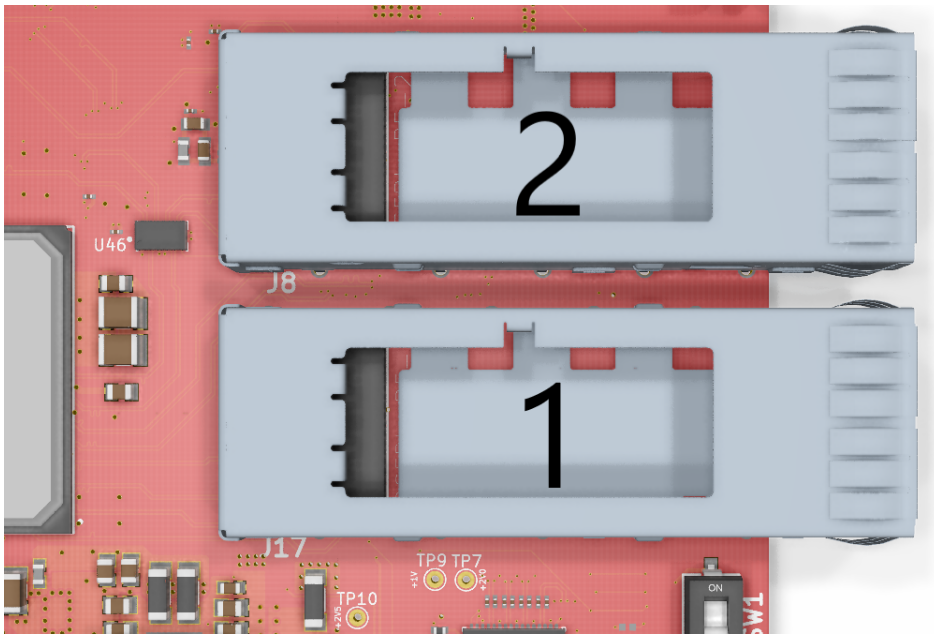


Figure 7: QSFP cages

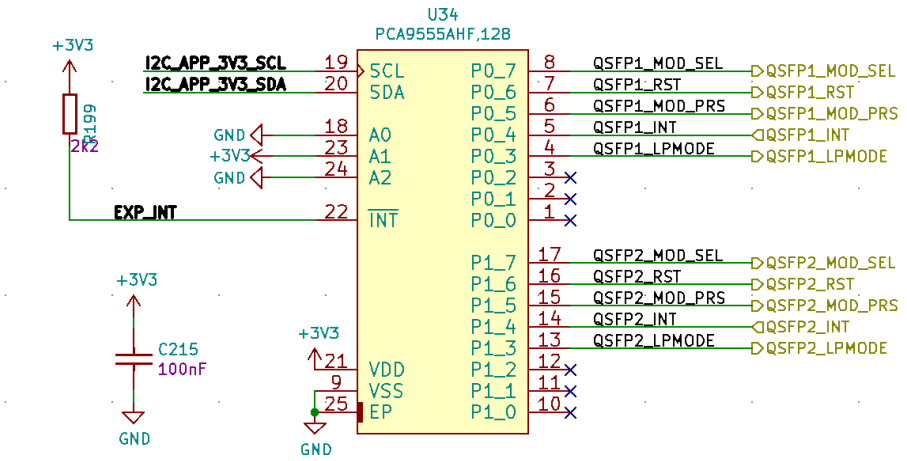


Figure 8: I2C IO expander which controls QSFPs signals

FPGA Pin	QSFP Signal
MGTXTXP0_116	QSFP1_TX_3_P
MGTXTXN0_116	QSFP1_TX_3_N
MGTXRXP0_116	QSFP1_RX_3_P
MGTXRXN0_116	QSFP1_RX_3_N
MGTXTXP1_116	QSFP1_TX_1_P
MGTXTXN1_116	QSFP1_TX_1_N
MGTXRXP1_116	QSFP1_RX_1_P
MGTXRXN1_116	QSFP1_RX_1_N
MGTXTXP2_116	QSFP1_TX_2_P
MGTXTXN2_116	QSFP1_TX_2_N
MGTXRXP2_116	QSFP1_RX_2_P
MGTXRXN2_116	QSFP1_RX_2_N
MGTXTXP3_116	QSFP1_TX_4_P
MGTXTXN3_116	QSFP1_TX_4_N
MGTXRXP3_116	QSFP1_RX_4_P
MGTXRXN3_116	QSFP1_RX_4_N

Table 7: QSFP1 pins connection

FPGA Pin	QSFP Signal
MGTXTXP0_115	QSFP1_TX_3_P
MGTXTXN0_115	QSFP1_TX_3_N
MGTXRXP0_115	QSFP1_RX_3_P
MGTXRXN0_115	QSFP1_RX_3_N
MGTXTXP1_115	QSFP1_TX_1_P
MGTXTXN1_115	QSFP1_TX_1_N
MGTXRXP1_115	QSFP1_RX_1_P
MGTXRXN1_115	QSFP1_RX_1_N
MGTXTXP2_115	QSFP1_TX_2_P
MGTXTXN2_115	QSFP1_TX_2_N
MGTXRXP2_115	QSFP1_RX_2_P
MGTXRXN2_115	QSFP1_RX_2_N
MGTXTXP3_115	QSFP1_TX_4_P
MGTXTXN3_115	QSFP1_TX_4_N
MGTXRXP3_115	QSFP1_RX_4_P
MGTXRXN3_115	QSFP1_RX_4_N

Table 8: QSFP2 pins connection

7 Ethernet

Marble is equipped with Ethernet PHY (88E1512) which supports 10/100/1000BASE-T. PHY is connected to the FPGA bank 16 via RGMII interface:

RGMII signal	FPGA pin
RGMII_RXD0	IO_L4N_16
RGMII_RXD1	IO_0_16
RGMII_RXD2	IO_L1P_16
RGMII_RXD3	IO_L1N_16
RGMII_RX_DV	IO_L4P_16
RGMII_RX_CLK	IO_L14P_16
RGMII_TXD0	IO_L6N_16
RGMII_TXD1	IO_L6P_16
RGMII_TXD2	IO_L8N_16
RGMII_TXD3	IO_L8P_16
RGMII_TX_EN	IO_L10P_16
RGMII_TX_CLK	IO_L11N_16
PHY_RSTn	IO_L10N_16

Table 9: RGMII pins assignment

The Ethernet PHY can be monitored over MDIO by the MMC. IP address and MAC address are stored in the MMC's internal EEPROM memory (EEPROM functionality is emulated in the internal flash). To read the internal PHY configuration registers, select the appropriate option from the MMC console. By default, the MDIO address is set to 1 but it can be changed by changing the resistor on the module. To set address to 1, desolder resistor R80 and solder resistor R65 with value 0R.

8 FMC

Both FMC sockets are equipped with HPC (High Pin Count) type connector but not all signals were connected to FPGA and MMC. The connection of both FMC connectors is shown below:

1. **FMC P1** signal connection:
   (a) LA00...LA33 - connected to FPGA banks 15 and 16.
   (b) HA00...HA23 - *not connected*.
   (c) HB00...HB21 - *not connected*.
   (d) CLK[0..1]_M2C - connected to FPGA bank 15.
   (e) GBTCLK[0..1]_M2C - connected to CLK MUX IN4 and IN5.
   (f) DP[0]_M2C - connected to high speed MUX.
   (g) DP[0]_C2M - connected to high speed MUX.
   (h) JTAG - connected to MMC.
   (i) I2C - connected to I2C bus shared with MMC and FPGA

2. **FMC P2** signal connection:
   (a) LA00...LA33 - connected to FPGA banks 12 and 14.
   (b) HA00...HA23 - connected to FPGA bank 13.
   (c) HB00...HB21 - *not connected*.
   (d) CLK[0..1]_M2C - connected to FPGA banks 12 and 14.
   (e) GBTCLK[0..1]_M2C - connected to CLK MUX IN6 and IN7.
   (f) DP[0..3]_M2C - connected to high speed MUX.
   (g) DP[0..3]_C2M - connected to high speed MUX.
   (h) JTAG - connected to MMC.
   (i) I2C - connected to I2C bus shared with MMC and FPGA

9 USB-UART

The USB-UART bridge (FT4232H) has 2 channels of UART:

9.1 FPGA UART

FPGA UART occupies FT4232's channel 3. UART signals are connected to bank 15 pins:

1. TX input IO_L1P_15
2. RX output IO_0_15

9.2 MMC UART

MMC UART occupies FT4232's channel 4 and is used as the MMC serial console. Terminal configuraton is 1N8 115200 baudrate. The FT4232's channel 2 DTR signal can be used to provide a reset signal for the microcontroller.

10 Power

Several hardware options are available to provide the nominal +12V power to the board. It can come from TE Connector (3-641119-2) - J19 (fig. 9) or by standard barrel connector (Type A: 5.5 mm OD, 2.1 mm ID) - J1. Input voltage range: 10V - 18V. Additionally, PoE can be used to power the module.

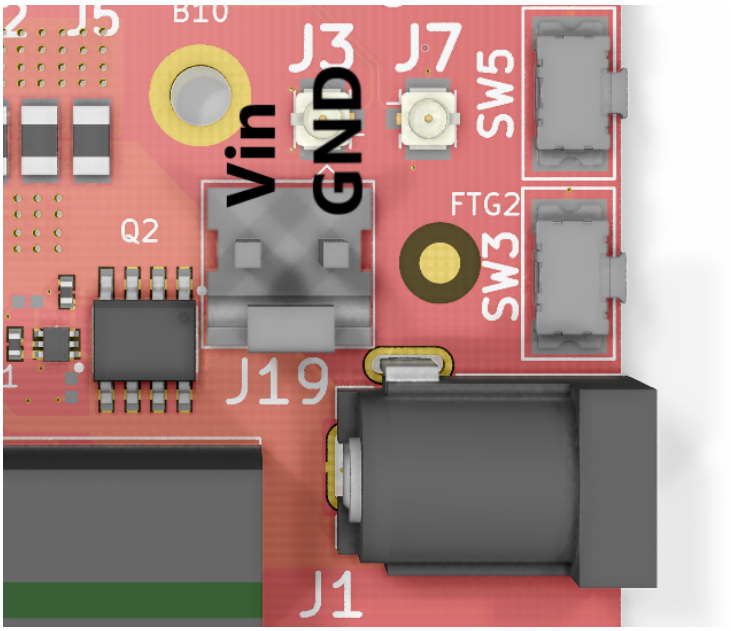


Figure 9: J19 connector with indicated Vin and GND

When the power is connected, the board starts up automatically. If there is a failure of any power rail, the LED corresponding to it will not light up.

Block diagram of Marble's power tree is shown in figure 10. The USB-UART bridge is powered from the micro USB connector via a converter (U22). The supervising microcontroller has a separate converter (U18) connected directly to the power input. The main voltages supplying the FPGA are produced by U35 - the programmable power management system. U35 turns on the individual power channels in the proper sequence and provides power to the additional converter (U58) and LDOs (U31, U36, U37, U47).

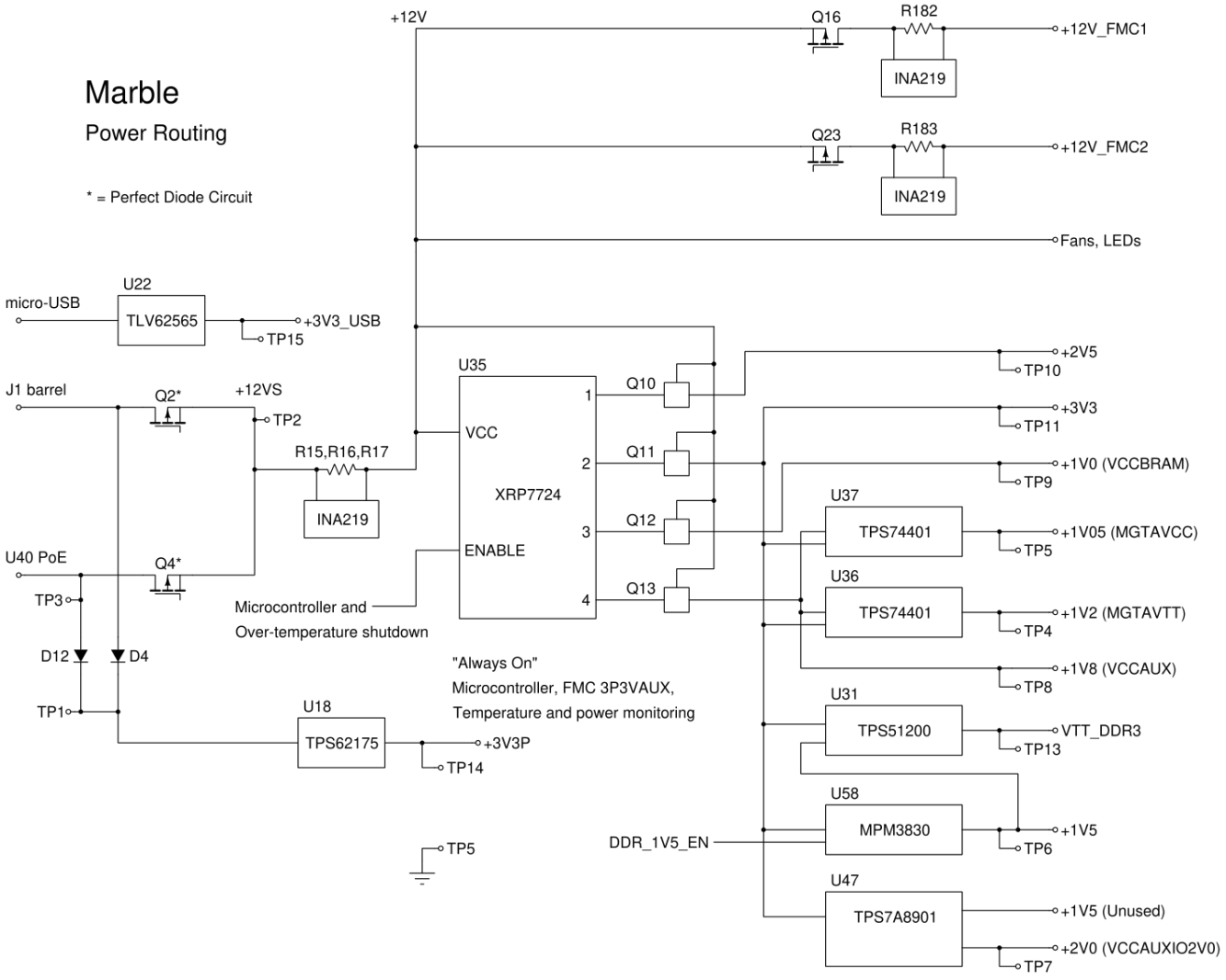


Figure 10: Marble power routing

Power supply features:

1. Over-temperature protection.
2. Power rails for the FPGA can be switched off and on by the microcontroller.
3. All power rails generated by XRP7724 can be monitored by the microcontroller and they are equipped with over-current protection.
4. Current consumption, voltage and rails status can be read by microcontroller.
5. The presence of the power rails is indicated by LED diodes
6. 12V power supply for both FMCs can be controlled independently by the microcontroller. Additionally, current can be measured.

10.1 Fan controller

The fan control and temperature monitoring of the FPGA chip is done with the MAX6639 chip. Selection, mounting, and configuration of the fans is outside the scope of this docu-

ment. The fans can be automatically controlled by measuring the temperature on the diode inside the FPGA. If the temperature exceeds a preset alarm threshold an ALERT signal will be issued. If the temperature continues to rise and exceeds another threshold, an "OVER-TEMP" signal will be issued and the FPGA will automatically power down. Through the I2C interface it is possible to read and write all MAX6639 configuration registers. Additionally, signals from both fan tachometers are monitored.

Two additional temperature sensors based on the LM75 chip provide temperature measurements around the main power converter and under the FMC P1 card. By default they are set to shut down the main power inverter when it exceeds 75 C.

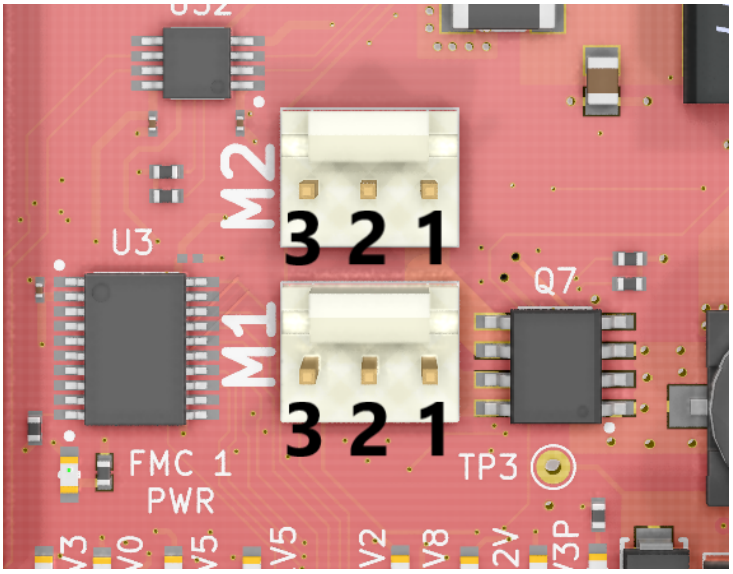


Figure 11: FANs connectors

	M1		M2
1	GND	1	GND
2	12V	2	12V
3	Tacho	3	Tacho

Table 10

11 MMC

Module Management Controller (MMC) is based on STM32F207 microcontroller and provides housekeeping functions such as:

1. Simple UART console over USB-UART bridge to control all functions
2. Monitoring voltage, current consumption and warning signals on power rails
3. Temperature monitoring at several locations

4. Controlling and monitoring fans
5. Configuring clock multiplexer
6. Configuring MGT switches
7. Resetting FPGA and controlling booting
8. Programming Power Management Controller
9. Controlling FMC power delivery and the presence of the cards
10. Ensuring communication over MDIO with Ethernet PHY
11. Ensuring communication over SPI with FPGA

11.1 Programing

MMC programing can be done by using external tools such as STM Nucleo-SWD programmer, SEGGER J-LINK Mini (Fig. 12, Fig. 13).

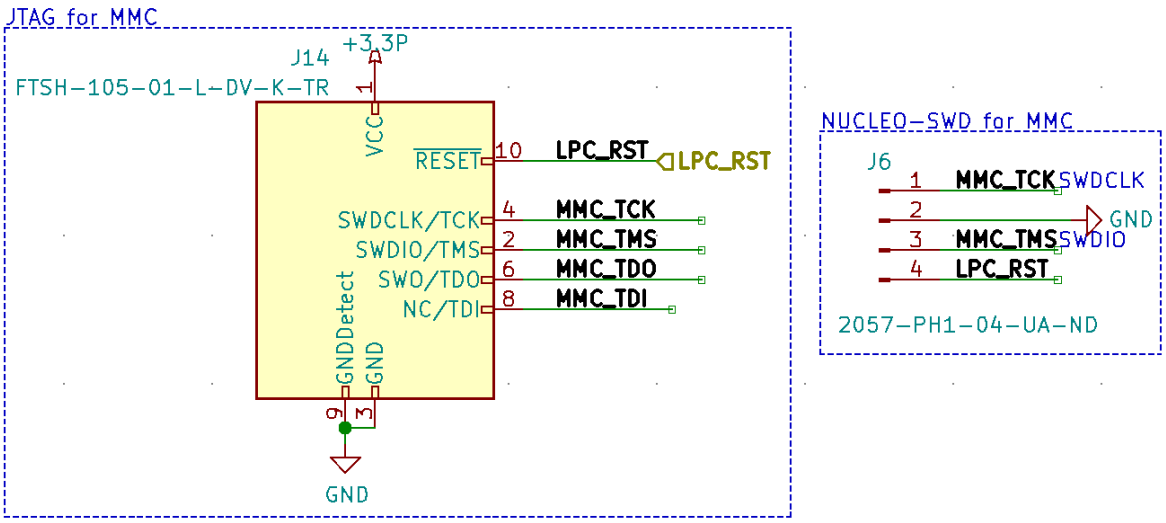


Figure 12: MMC JTAG and SWD interfaces

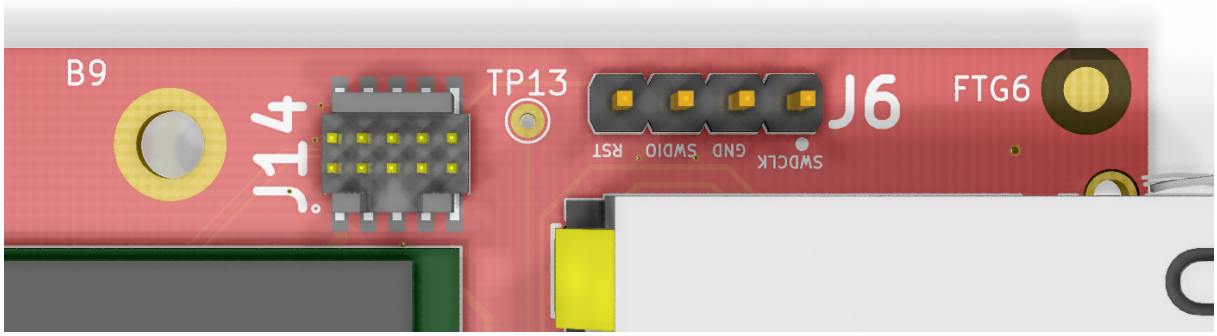


Figure 13: MMC JTAG and SWD connectors

Download the latest version of the microcontroller testing code from GitHub:

https://https://gitlab.lbl.gov/spaiagua/marble_mmc/-/tree/unified_marble

A recent version of OpenOCD (v0.10.0 or later) is required.

1. Connect JTAG module to **J14**
2. Connect the micro USB cable and using the serial terminal, connect to the last serial port for the new listed in the operating system. Use 115200 baudrate.
3. Power up the board.
4. Program the microcontroller using the following commands:
   (a) Go to the main folder of the downloaded repository.
   (b) Open command terminal and run command:

```
$ make marble_download
```

5. After successful programming, a menu in the serial terminal should appeared and LEDs (LD15, LD11, LD12) should blink in the "snake" pattern.

11.2 LEDs

Three general purpose LEDs are connected to the MMC chip:

1. LED11 - connected to pin PE1
2. LED12 - connected to pin PE2
3. LED15 - connected to pin PE0

11.3 I2C Tree

Marble is equipped with two I2C buses (block diagram is shown in fig. 14):

1. I2C_PM - supports devices for power management, temperature measurement and fan control
2. I2C_FPGA - This bus is shared between the MMC and the FPGA chip. Through an I2C switch, it is connected to:
   (a) FMC 1 and FMC 2
   (b) Clock multiplexer
   (c) SO-DIMM module
   (d) QSFP 1 and QSFP 2
   (e) Current measurement devices, Si570 and IO expanders

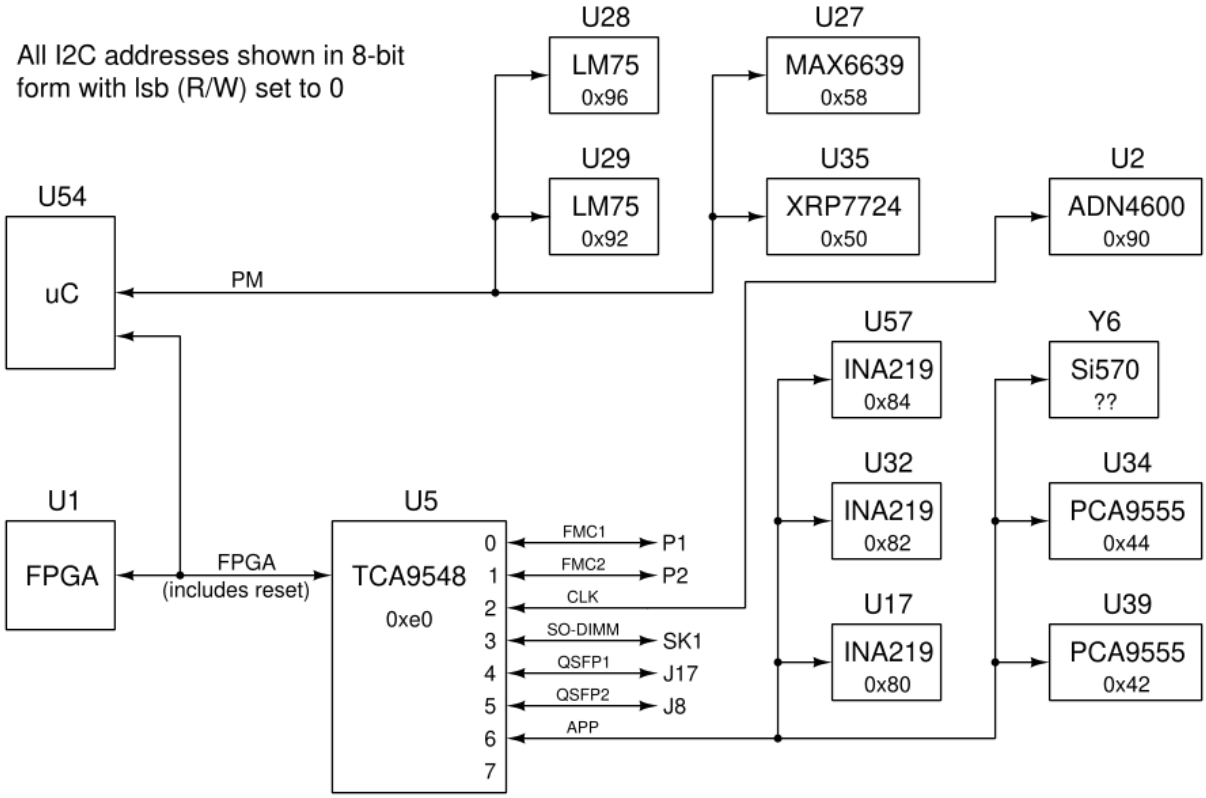


Figure 14: I2C map

12 Mechanical dimensions

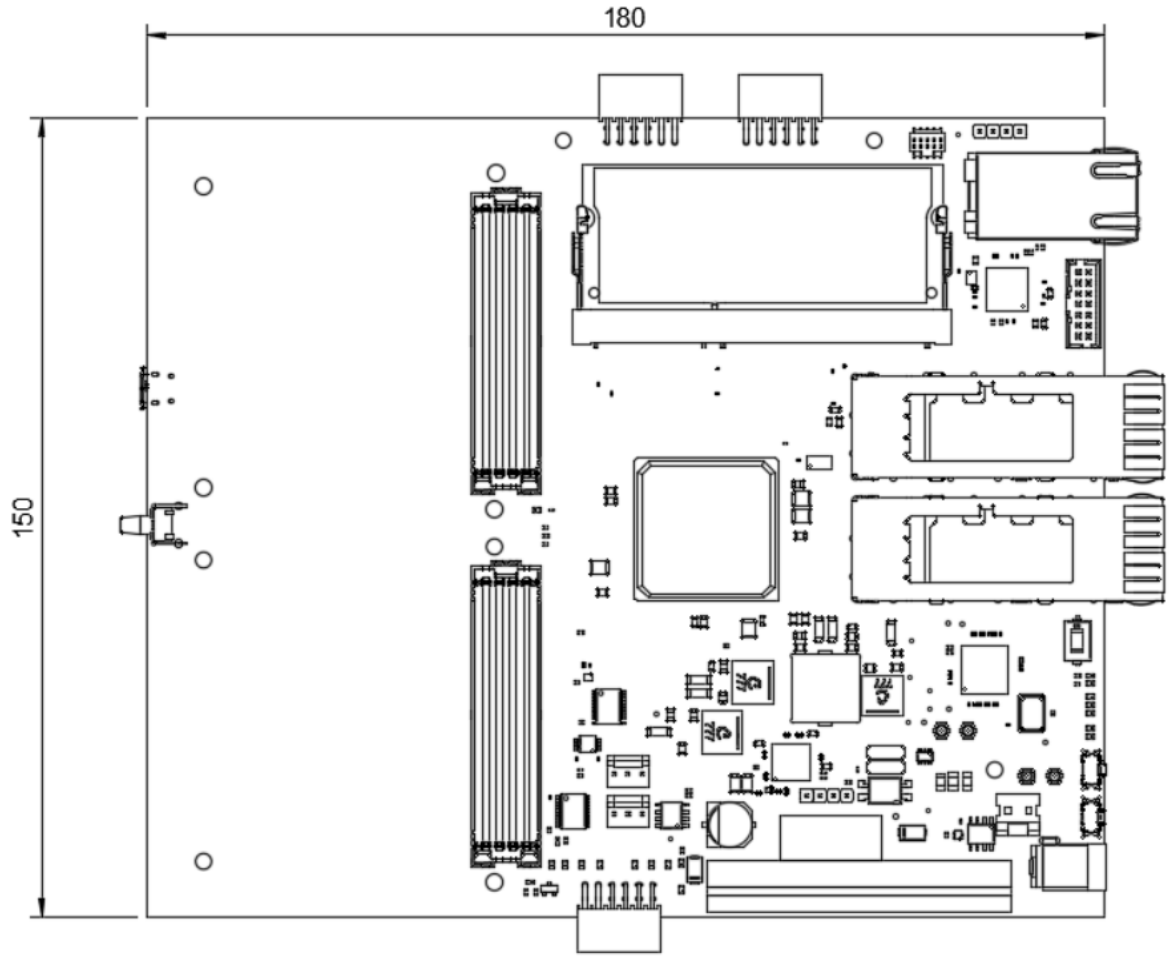


Figure 15: Mechanical dimensions

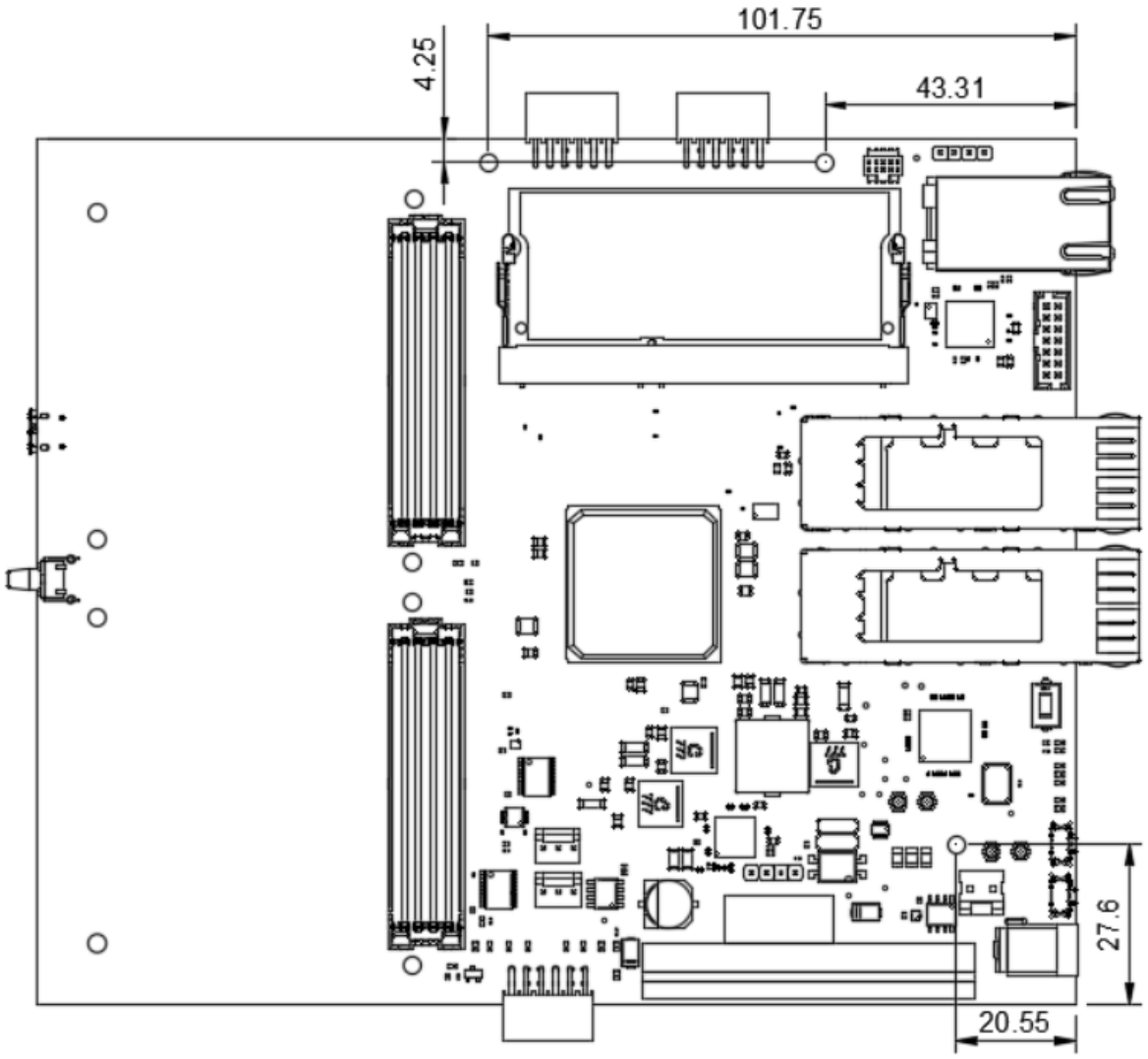


Figure 16: Mounting holes positioning

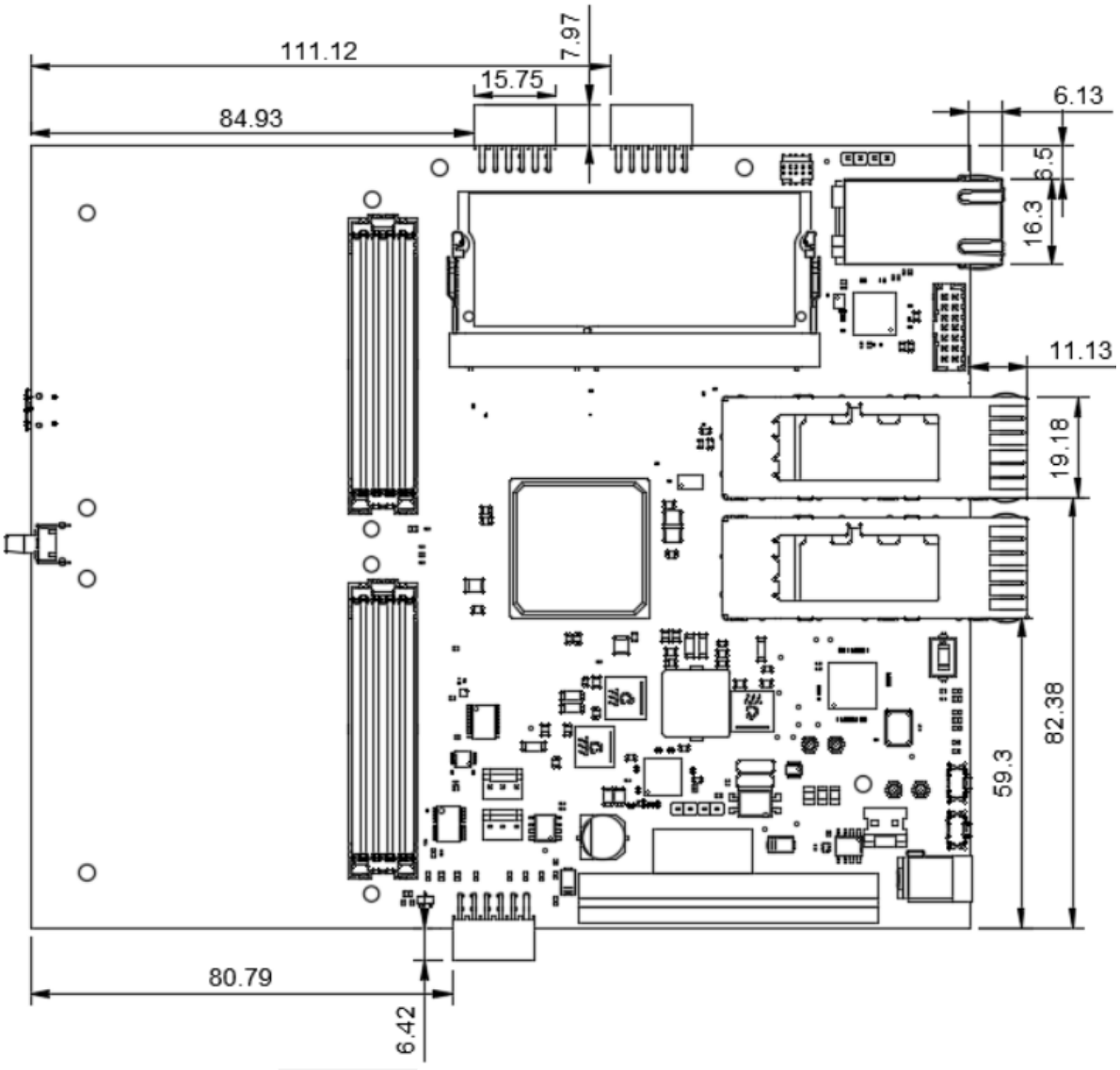


Figure 17: Protruding connectors positioning

13 Appendix

13.1 Power supply test points

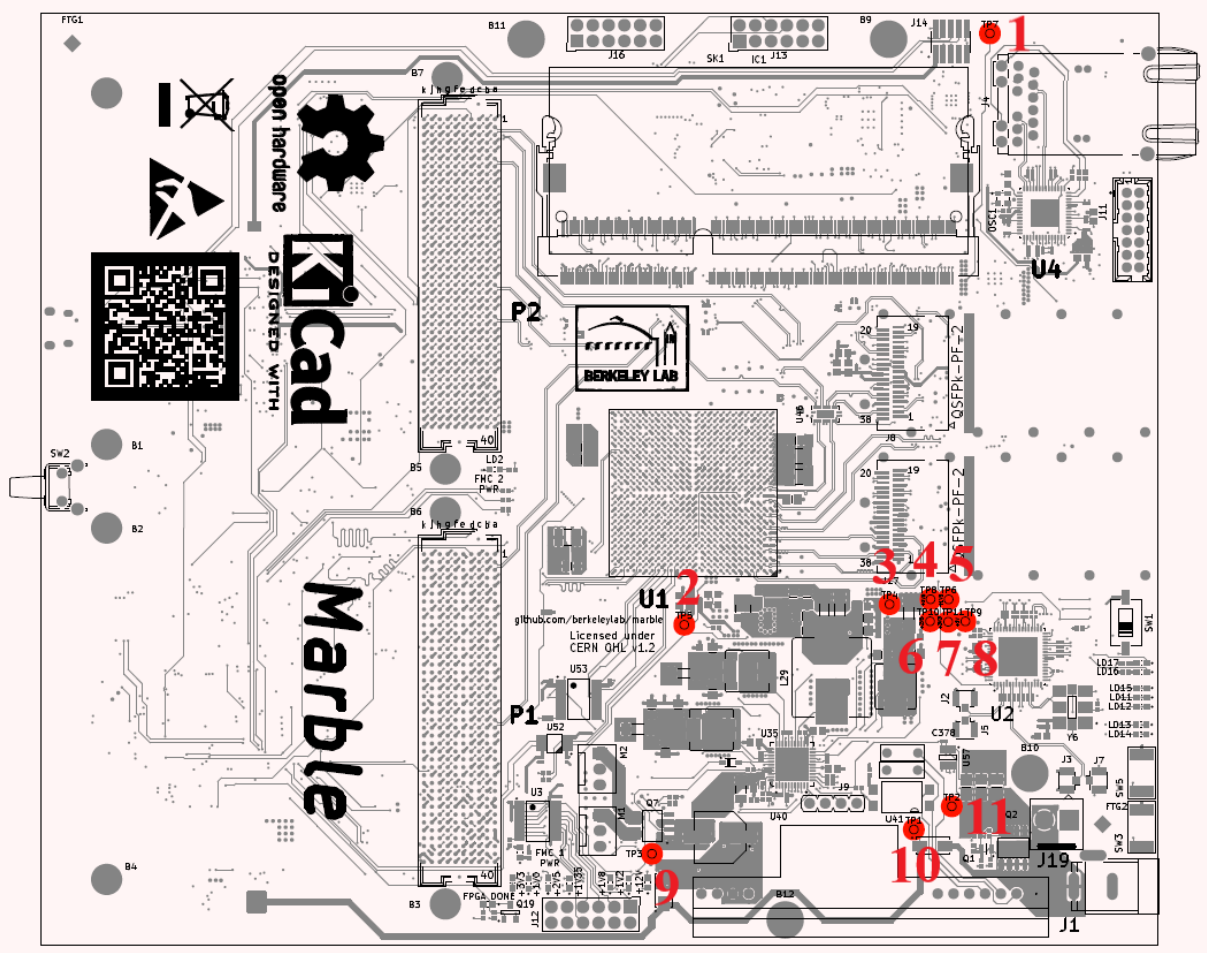


Figure 18: Test point map

(a) Number 1 - **TP13 VTT_DDR3 (0.75V).**

(b) Number 2 - **TP5 MGTAVCC_DDR3 (1.05V).**

(c) Number 3 - **PoE**

(d) Number 4 - **MGTAVTT (+1.2V)**

(e) Number 5 - **MGTAVCC (+1.05V)**

(f) Number 6 - **+1V5**

(g) Number 7 - **VCCAUXIO2V0 (+2.0V)**

(h) Number 8 - **VCCAUX (+1.8V)**

(i) Number 9 - **VCCBRAM (+1.0V)**

(j) Number **10 - +2V5**

(k) Number **11 - +3V3**

(l) Number **12 - GND**

(m) Number **13 - VTT_DDR3 (+0.75V)**

(n) Number **14 - +3.3P (always-on)**

(o) Number **15 - +3V3_USB**

Test points 12-15 are not present on Marble v1.0.

References